

Temperature Compensation Asymmetry at Thermal Extremes in 3-D NAND Flash: Physical Mechanisms and State Dependence

Yicheng Ke^{ID}, Minghui Dong^{ID}, Linlin Cai^{ID}, *Member, IEEE*, and Wangyong Chen^{ID}, *Member, IEEE*

Abstract—This work experimentally isolates the effective behavior of implemented on-chip temperature compensation (TC) in commercial TLC charge-trapping 3-D NAND using paired distribution-level measurements with TC and without TC in the same readout domain. We evaluate threshold-voltage (V_{TH}) statistics of program states P1–P7 from -55°C to 125°C under high-temperature-program/low-temperature-read (HPLR) and low-temperature-program/high-temperature-read (LPHR) trajectories. Although TC stabilizes the mean V_{TH} within the nominal window (0°C – 70°C), it exhibits a systematic endpoint mismatch outside this range, manifested as undercompensation at low (UCL) temperatures and overcompensation at high (OCH) temperatures. OCH is concentrated in high program states (P5–P7), compressing adjacent-state spacing and reducing the effective read window. By comparing the paired with TC and without TC statistics, we show that the implemented TC correction is mainly temperature dependent and only weakly state dependent, whereas the intrinsic $\Delta V_{TH}(T)$ response is nonlinear, trajectory dependent, and strongly state dependent. Technology computer-aided design (TCAD) simulations further link this mismatch to temperature-regime-dependent charge behavior, including shallow/deep trap-energy occupancy, rate-limited trapping/detrapping and redistribution, Poole–Frenkel (PF)-assisted emission, and mobility degradation under large stored charge. These results distinguish compensation-induced mismatch from intrinsic device drift and motivate state-aware, temperature-region-aware TC calibration for wide-temperature 3-D NAND read robustness.

Index Terms—3-D NAND flash, technology computer-aided design (TCAD), temperature compensation (TC), wide-temperature.

I. INTRODUCTION

THREE-DIMENSIONAL 3-D NAND flash memory has established itself as the dominant nonvolatile storage solution, driven by its superior density and cost efficiency

Received 14 May 2026; accepted 24 June 2026. Date of publication 2 July 2026; date of current version 5 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62304263, in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2024A1515010349, and in part by under Grant 2024-JCJQ-JJ-0220. The review of this article was arranged by Editor B. J. O’Sullivan. (Corresponding authors: Wangyong Chen; Linlin Cai.)

The authors are with the School of Microelectronics Science and Technology, Sun Yat-Sen University, Guangzhou 510275, China, and also with the Zhuhai Key Laboratory of Intelligent EDA and Reliability Design of Advanced Integrated Circuits, Zhuhai 519082, China (e-mail: caillin3@mail.sysu.edu.cn; chenwangyong@mail.sysu.edu.cn).

Digital Object Identifier 10.1109/TED.2026.3708111

[1]. As the technology continues to scale in layer count and wordline pitch, ensuring read reliability becomes increasingly challenging. The integrity of data retrieval is fundamentally dictated by the tightness of threshold-voltage (V_{TH}) distributions and the guard-band margins between multilevel states [2], [3], [4]. Among various reliability detractors, temperature variation poses a critical threat because it alters charge-trapping dynamics and read sensing conditions, thereby inducing V_{TH} drift and distribution distortion that jeopardize multilevel sensing accuracy [2], [4], [5], [6].

To mitigate these thermal effects, commercial 3-D NAND products widely integrate on-chip temperature compensation (TC), which dynamically adjusts sensing reference voltages based on precharacterized temperature coefficients to stabilize read margins [7], [8], [9]. From an application perspective, operating-temperature requirements vary significantly across domains. While standard applications operate within specific temperature windows, emerging aerospace and other mission-critical applications further extend the required coverage toward wider and more extreme-temperature ranges. Fig. 1(a) summarizes representative temperature windows across application domains and motivates the necessity of understanding TC behavior under extreme temperatures. From a device-statistics perspective, TC is expected to regulate distribution alignment by shifting sensing references so that state distributions remain properly separated across temperature. Fig. 1(b) illustrates how TC aligns temperature-dependent V_{TH} distributions to preserve the effective read window.

However, conventional TC is typically implemented as a simplified, state-insensitive linear correction with a unified slope. This one-size-fits-all approach assumes a uniform thermal response across program states, which is not robust for charge-trapping devices. As the program level increases, the stored-charge density and the vertical-field distribution change markedly, so the intrinsic, state-dependent thermal response can deviate from the applied linear correction, especially at temperature extremes. Such a device–TC mismatch not only leaves residual drift but can also perturb the intended interstate separation, manifesting as read-window loss at the distribution level.

In this work, we conduct a distribution-level assessment of on-chip TC in charge-trapping 3-D NAND over an extreme range from -55°C to 125°C , and we expose a two-end mismatch evaluated against the ideal zero-drift TC target, defined as a temperature-invariant mean V_{TH} in the readout domain. Deviations from this target appear as

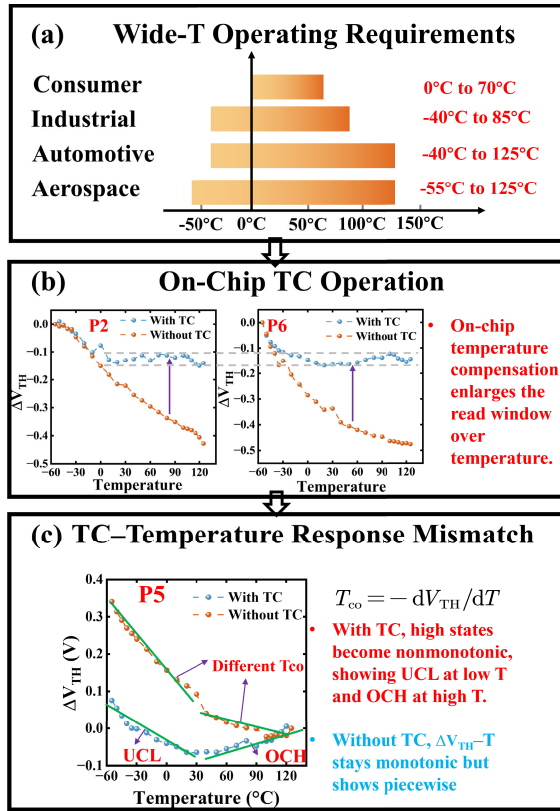


Fig. 1. Distribution-centric evaluation of on-chip TC in charge-trapping 3-D NAND. (a) Representative operating-temperature requirements across application domains. (b) TC realigns V_{TH} distributions to preserve the read window across temperature. (c) Extreme-temperature mismatch of linear TC, showing UCL temperatures and OCH temperatures, with stronger OCH in high program states. $\Delta V_{TH}(T) = V_{TH}(T) - V_{TH}(T_{prog})$.

undercompensation at low (UCL) temperatures and overcompensation at high (OCH) temperatures. Fig. 1(c) summarizes the observed UCL/OCH behaviors and shows that the OCH impact becomes more pronounced in high program states, where interstate spacing is compressed, and the effective read window is reduced. Fig. 2 compares $\Delta V_{TH}(T)$ with TC and without TC along two temperature trajectories. High-temperature-program/low-temperature-read (HPLR) denotes the high-temperature program and low-temperature read trajectory, whereas low-temperature-program/high-temperature-read (LPHR) denotes the low-temperature program and high-temperature read trajectory. In particular, LPHR emulates a cold-start programming condition followed by readout under device self-heating, while HPLR emulates programming during sustained high-performance operation at elevated temperature followed by readout after power-off cooling, thereby capturing practical worst case use scenarios.

II. EXTREME-TEMPERATURE CHARACTERIZATION AND CROSS-TEMPERATURE TRAJECTORIES

The goal of this experiment is to diagnose endpoint mismatch of on-chip TC under extreme temperatures, with a particular focus on UCL and OCH. We therefore performed a wide-temperature characterization from -55°C to 125°C on commercial charge-trapping 232-layer 3-D NAND chips

operated in TLC mode, beyond the nominal 0°C – 70°C operating window specified by the vendor [5], [10], [11], [12]. The tested device follows a vertical-channel charge-trapping NAND architecture with a lightly doped polysilicon channel and an oxide/nitride/oxide stack. All tested devices were fresh samples selected from the same manufacturing lot, comprising three chips with two dies evaluated per chip. For each trajectory, measurements were repeated with TC and without TC using an identical readout flow, forming paired datasets that isolate compensation artifacts from the intrinsic device response. This paired on-chip protocol enables the implemented TC action to be evaluated directly in the same readout domain, rather than inferring compensation behavior only from TC-enabled drift or from an external correction model.

To avoid state/pattern mixing, each die used a state-dedicated solid pattern with seven independent blocks assigned to P1–P7, as shown in Fig. 3(a). The temperature sweep plan and two cross-temperature trajectories are summarized in Fig. 3(b). At each temperature point, each state block contributed 90 sampled pages, evenly split among the top, middle, and bottom WL regions, and the same sampling scheme was used for all states, trajectories, and TC modes. The reported descriptors were averaged over the sampled pages and measured dies. Measurement uncertainty was quantified by page-to-page and die-to-die relative spreads under matched program state, temperature point, trajectory, and WL-region conditions. Across P1–P7, the page-to-page relative spread is 0.88%–3.73% under LPHR and 1.11%–4.53% under HPLR, while the die-to-die relative spread is 1.22%–5.40% and 1.31%–5.23%, respectively, confirming that the TC-induced high-state mismatch is not dominated by page-level or die-level fluctuation. Programming was performed at -55°C and 125°C for low- and high-temperature anchors, respectively. After programming, readout was carried out along the designated paths on a thermal-shock platform. A dwell period was inserted before each sweep to suppress early postprogram relaxation [13], [14]. A retention-control check after the same temperature-transition procedure showed only mV-level post-transition mean drift, indicating that retention-related drift is minor under the present fixed protocol. All reads were taken after thermal settling using a fixed sensing criterion.

Fig. 3(c) outlines the distribution reconstruction procedure. For each sampled page, V_{ref} was swept and raw bit error counts were collected, and count differences between adjacent V_{ref} levels were used to reconstruct histogram-form V_{TH} distributions, as shown in Fig. 3(d) [15]. The distribution mean was extracted using one fixed estimator across all temperatures, states, and TC modes, thereby keeping the comparison in a consistent readout domain. Accordingly, the mean shift is defined as

$$\Delta V_{TH}(T, P) = V_{TH}(T, P) - V_{TH}(T_{pgm}, P). \quad (1)$$

For measurements with TC and without TC, ΔV_{TH} is calculated separately using the same readout-domain extraction protocol. The extracted $V_{TH}(T)$ and reconstructed distributions are then used for fitting and for quantifying the TC endpoint mismatch, with emphasis on UCL and OCH. Fig. 3(d) presents

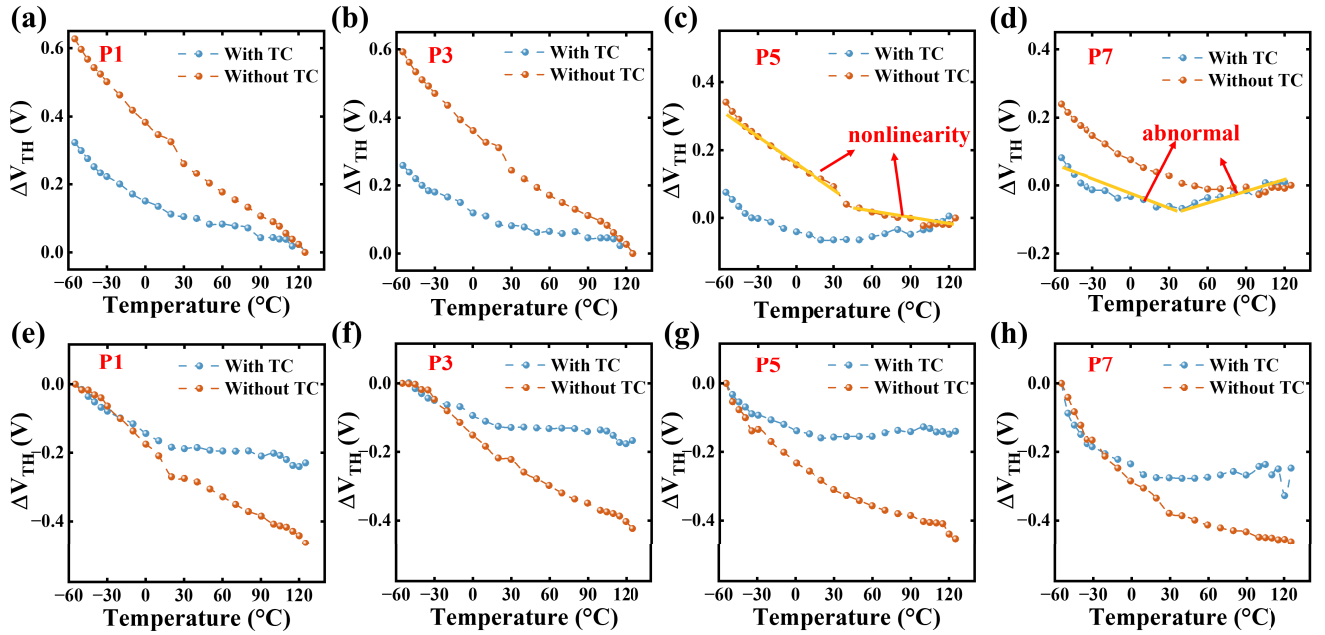


Fig. 2. Temperature dependence of the distribution-mean $\Delta V_{TH}(T)$ for representative TLC states (P1, P3, P5, and P7) with TC and without TC. (a)–(d) and (e)–(h) HPLR and LPHR trajectories, respectively. Each panel compares the responses with TC and without TC responses in the same readout domain.

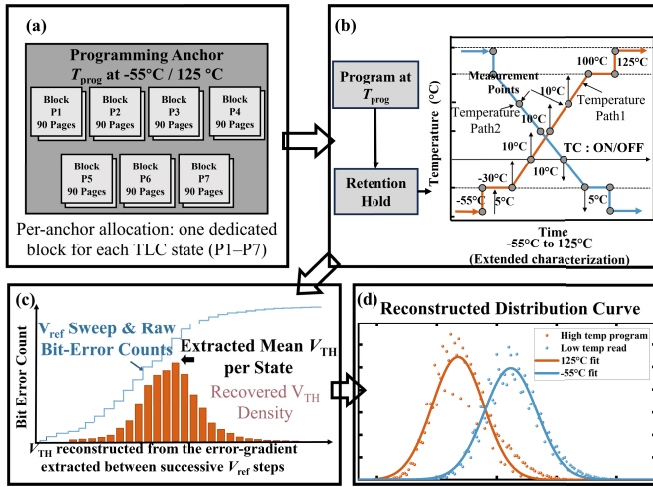


Fig. 3. Methodology for cross-temperature characterization and distribution level V_{TH} extraction. (a) Dedicated block mapping for state programming from P1 to P7. (b) Temperature sweep schedule and defined thermal paths. (c) Distribution recovery by stepping the read reference voltage V_{ref} and collecting raw bit error counts. (d) Representative reconstructed V_{TH} distribution measured under HPLR.

a representative extracted V_{TH} distribution, which is subsequently used for fitting and quantitative evaluation.

III. RESULTS AND DISCUSSION

A. State–Temperature Mapping of Mean V_{TH} Drift With and Without TC

As illustrated in Fig. 2, (a)–(d) correspond to the HPLR trajectory, whereas (e)–(h) correspond to the LPHR trajectory. To highlight the key behaviors while maintaining a consistent analysis framework [5], [10], we select four representative

states (P1, P3, P5, and P7) and compare their $\Delta V_{TH}(T)$ with TC and without TC. Under HPLR, P1 and P3 remain well regulated with TC within the nominal operating window, exhibiting a flattened and stable response. Under HPLR, high states (P5–P7) exhibit OCH, including a slope reversal relative to the baseline without TC. Under LPHR, the behavior with TC is overall better controlled, and no strong inversion is observed. A pervasive UCL is still present near the low-temperature endpoint, manifested as residual drift under TC due to insufficient correction. These with TC trends are also preserved in the top, middle, and bottom WL-region groups. Although region-dependent offsets exist, the main UCL/OCH characteristics remain consistent across the sampled WL regions, indicating that the observed behavior cannot be attributed to one specific WL-region group.

Because the circuit-level implementation of the commercial on-chip TC is not accessible, we evaluate its behavior through the effective distribution-mean correction shown in Fig. 4. Fig. 4(a) and (b) shows the map of this correction under the HPLR and LPHR trajectories, respectively, and they reveal the same key pattern: the correction is dominated by temperature and shows only weak dependence on program state. Weak program-state dependence refers to the extracted TC correction map, rather than to the measured $V_{TH}(T)$ response. The applied adjustment remains small at the cold end and increases steadily toward the hot end, indicating a stronger reference shift at elevated temperature. This weak state selectivity is largely acceptable for low and mid states, but it becomes a primary limiter once the intrinsic mean V_{TH} response turns strongly state dependent at high temperature under large stored charge. In that regime, a single-slope, near state-independent correction cannot track the high-state mean evolution and therefore biases P5–P7 toward OCH, consistent with the

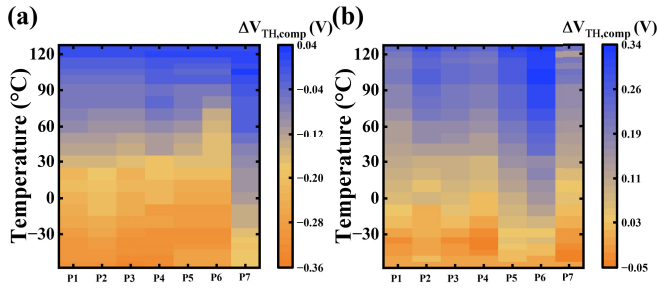


Fig. 4. Mean-level TC compensation mapped over temperature and program state. The effective correction is defined as $\Delta V_{TH,comp}(T, P) = V_{TH}^{with\ TC}(T, P) - V_{TH}^{without\ TC}(T, P)$ and visualized as a heatmap across P1–P7. (a) and (b) HPLR and LPHR trajectories, respectively.

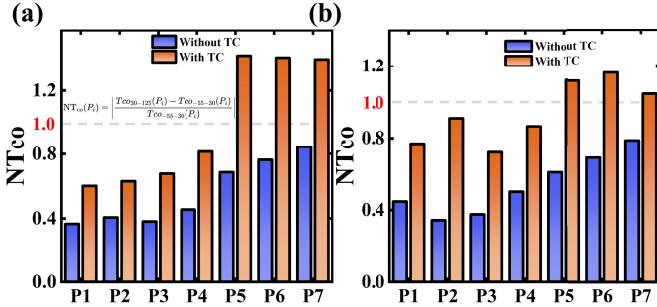


Fig. 5. Normalized temperature-coefficient metric (NT_{co}) obtained from a two-region slope evaluation of $V_{TH}(T)$ and compared with TC and without TC conditions across program states P1–P7. (a) and (b) HPLR and LPHR trajectories, respectively.

observed high-temperature anomaly [7]. Fig. 2 shows that $\Delta V_{TH}(T)$ changes much more steeply in the low-temperature region, and applying one linear TC slope over the full temperature span therefore leaves insufficient correction at low temperature, naturally manifesting as UCL.

While the $\Delta V_{TH}-T$ curves show the detailed temperature trajectory of each program state, the normalized temperature coefficient (NT_{co}) is used here as a compact indicator of low-/high-temperature slope asymmetry. To quantify this contrast, NT_{co} is defined in (2) using segmented slopes of $V_{TH}(T)$, where $T_{co,30-125}$ and $T_{co,-55-30}$ represent the extracted temperature coefficients (slopes of V_{TH} versus T) in the high-temperature (30 °C–125 °C) and low-temperature (from –55 °C to 30 °C) regions, respectively. Fig. 5(a) and (b) shows the report NT_{co} under HPLR and LPHR, respectively. With TC, NT_{co} increases across states relative to the baseline without TC, and the increase concentrates in the high program region. Notably, NT_{co} exceeds 1 for P5 through P7, indicating that the with TC mean response becomes markedly more asymmetric between the low- and high-temperature regions in these high states. This identifies P5–P7 as the main states where the effective TC correction departs from the without TC state-dependent thermal response, consistent with the high-temperature mean-drift inversion and the observed OCH

$$NT_{co}(P_i) = \left| \frac{T_{co,30-125}(P_i) - T_{co,-55-30}(P_i)}{T_{co,-55-30}(P_i)} \right|. \quad (2)$$

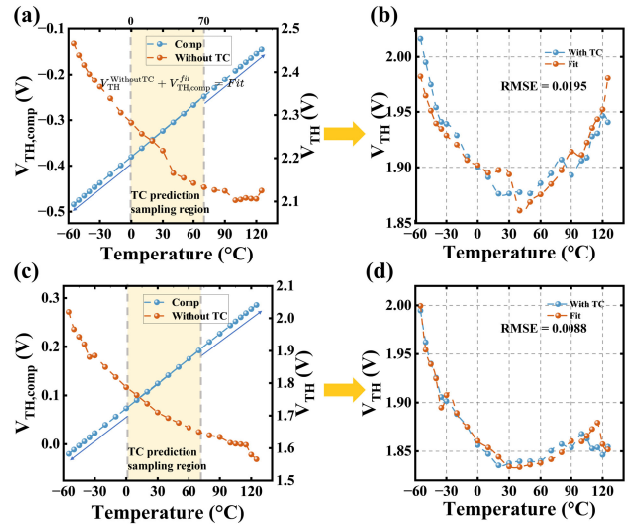


Fig. 6. Linear TC emulation and distribution-level validation. (a) Without TC V_{TH} distribution with the calibrated linear compensation function overlaid for HPLR. (b) Replayed distribution after applying the calibrated correction compared with the measured with TC distribution for HPLR. (c) and (d) Same calibration-and-replay validation performed for LPHR.

B. Validating the Linear, State-Insensitive TC Model

Fig. 6 shows the validation that the on-chip TC can be emulated by an equivalent single-slope linear correction. In this replay analysis, the without TC distribution is used as the uncompensated reference, and a linear correction is then applied to reproduce the measured with TC response in the same readout domain. For the HPLR trajectory, Fig. 6(a) shows the without TC V_{TH} distribution together with a calibrated linear compensation function obtained from the measured temperature-dependent shift. The calibrated correction is then applied to the without TC distribution to generate a “TC-replayed” distribution. Fig. 6(b) compares this replayed result with the measured TC distribution, and the close overlap confirms that the TC action is well captured by the linear model. The same calibration-and-replay procedure is performed for the LPHR trajectory in Fig. 6(c) and (d), where the agreement remains strong, indicating that the linear emulation holds for both cross-temperature paths.

Fig. 7 shows the further checks whether the correction is state dependent by enforcing a unified compensation slope across all seven program states under HPLR and plotting the resulting $V_{TH}(T)$ responses. The intercept offsets mainly arise from the state-dependent baseline V_{TH} established by ISPP programming, because different program states contain different stored-charge levels, vertical-field conditions, and trap-occupation configurations. In the measurements with TC, these offsets may also include residual alignment errors associated with the calibration anchor of the on-chip TC. Across P1–P7, the extracted slopes remain consistent, indicating that the implemented TC primarily applies a state-insensitive linear slope.

Finally, we attribute part of the observed UCL and OCH behavior to how TC is calibrated in practice. Since the primary objective is to protect the nominal operating window (0 °C–70 °C), the compensation curve is likely optimized

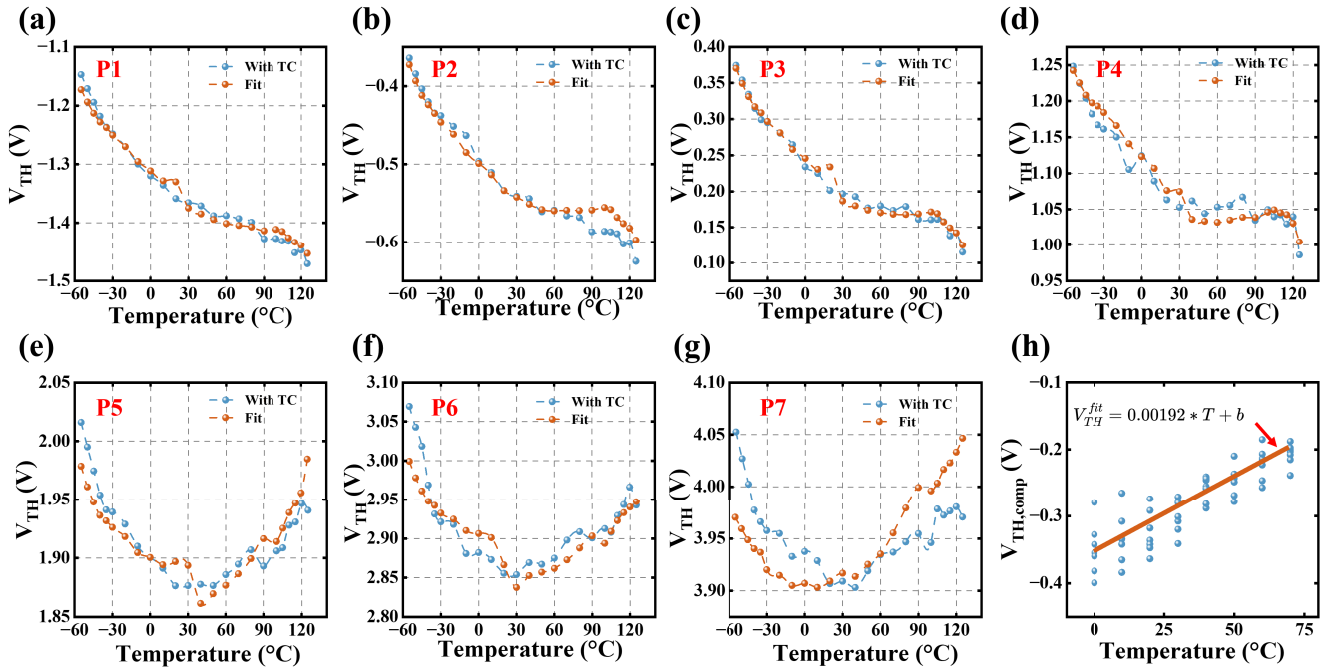


Fig. 7. State-insensitive slope verification of the implemented TC under HPLR. (a)–(g) $V_{TH}(T)$ responses for P1–P7 after applying a unified compensation slope. (h) Window-prioritized calibration using 0 °C–70 °C points and global replay over the full temperature span.

using data within this range. As shown in Fig. 7(h), we fit the TC slope using only the 0 °C–70 °C compensation points and then replay the resulting linear correction over the full temperature span to assess endpoint behavior. This window-prioritized calibration reproduces the measured trends, yielding UCL and OCH. After establishing the effective unified-slope behavior of the implemented TC, we further quantify its read-window impact using the adjacent-state read margin [16]. The margin is defined as

$$RM_i(T) = [V_{TH}(T, P_{i+1}) - k\sigma_{i+1}(T)] - [V_{TH}(T, P_i) + k\sigma_i(T)] \quad (3)$$

where V_{TH} and σ are extracted from the reconstructed state distributions, and k is the distribution-boundary coefficient. In this work, $k = 3$ is used as a 3σ boundary criterion to estimate the effective separation between adjacent-state distributions. A smaller RM_i indicates a reduced adjacent-state read window, and a negative value indicates overlap of the selected distribution boundaries. As shown in Fig. 8, the read-margin curves directly show how adjacent-state separation evolves along each cross-temperature trajectory. The label “mild RM loss” denotes regions where the magnitude or rate of read-margin reduction is relatively small. These results show that the compensation mismatch and distribution broadening can translate into measurable adjacent-state read-margin degradation at the distribution level.

C. Mechanism Interpretation and TCAD Corroboration

To explain the observed UCL and the high-state OCH, we attribute the mean $V_{TH}(T)$ response to a transition of dominant mechanisms from low to high temperature, with

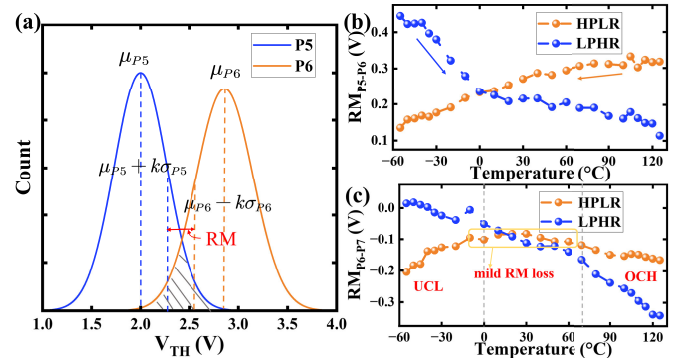


Fig. 8. Distribution-level adjacent-state read-margin evaluation under the implemented TC. (a) Schematic definition of adjacent-state read margin using the reconstructed distribution mean and width. (b) P5–P6 and (c) P6–P7 read-margin evolution under HPLR and LPHR with $k = 3$. Arrows indicate the imposed cross-temperature direction.

cross-temperature trajectories shaping the trap-energy occupancy and with trap relaxation and charge trapping/detrapping dynamics playing the dominant role, while temperature-dependent polysilicon-channel mobility and bandgap narrowing (BGN) act as secondary readout-domain effects. In the low-temperature regime illustrated in Fig. 9(a), thermally activated emission and trap-assisted transport are strongly suppressed, so the trapped-charge configuration is governed by shallow/deep trap-energy occupancy and rate-limited redistribution during the temperature transition, producing a steep low-temperature $V_{TH}(T)$ response [6], [17], [18]. Because practical on-chip TC is typically optimized for the nominal operating window where this effect is much weaker, a single global linear slope cannot match both the steep cold-end behavior and the mild in-window behavior, naturally leading to UCL.

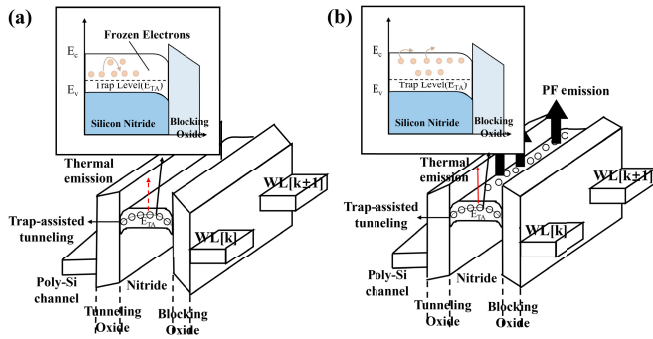


Fig. 9. (a) Dominant V_{TH} variation mechanisms at low temperature. (b) Dominant V_{TH} variation mechanisms at high temperature.

At high temperature, the dominant processes shift to those in Fig. 9(b). Poole–Frenkel (PF) emission under strong vertical field accelerates charge emission and redistribution, while channel mobility degradation increases the temperature sensitivity of the read condition [6], [19], [20]. In high program states, the larger stored charge produces a stronger vertical field and thus more severe mobility degradation, amplifying the intrinsic state dependence of the mean response. As a result, a state-insensitive, unified-slope correction cannot track the high-state evolution and is prone to OCH, which drives the observed high-state OCH.

These interpretations are supported by Sentaurus technology computer-aided design (TCAD) simulations of a vertical-channel charge-trapping 3-D NAND cell. Fig. 10(a) shows the simulated structure, including the silicon channel, tunnel oxide, charge-trapping nitride, blocking oxide, and TiN gate electrodes. The key layer/WL dimensions are parametrically defined, with a $1 \times 10^{15} \text{ cm}^{-3}$ p-type channel and $1 \times 10^{20} \text{ cm}^{-3}$ n-type source/drain. The simulations include PF emission, nonlocal barrier tunneling, and temperature-dependent mobility. Cross-temperature operation is emulated by temperature ramping between anchors with traps active during the ramp and frozen during the subsequent hold and read to preserve a readout-matched domain [18]. We introduce η as an effective trap-evolution parameter during the temperature transition. It is defined on the TCAD transient axis and used to modulate the effective rate of trap relaxation, charge emission, and spatial redistribution. By varying η , we examine the sensitivity of the simulated V_{TH} turning behavior to the trap-evolution time scale, rather than quantitatively reproducing the actual thermal-shock ramp or package-level stabilization process.

Building on the regime-level picture in Fig. 9, the simulated $\Delta V_{TH}(T)$ results in Fig. 10(b)–(i) reveal a clear trajectory-dependent sensitivity to the trap-evolution setting η , and Fig. 11 shows the further anchors of the turning temperature with consistent internal signatures at a representative η . Under HPLR, the turning point shows weak dependence on η , consistent with a charge configuration dominated by deeper traps, making the turning point less sensitive to the trap-evolution setting. Under LPHR, the turning point shifts toward lower temperature as η increases, indicating a larger shallow or nonequilibrium component whose evolution

is rate-limited during the heating transition. This deep-versus-shallow occupancy contrast also explains the different high-temperature baselines without TC: HPLR high states tend to flatten at the hot end due to deep-trap dominance, whereas LPHR high states can continue to decrease because shallow components remain active. Consequently, when a unified linear TC slope is applied, the stronger hot-end flattening in HPLR makes high states more susceptible to OCH and a clear inversion, while LPHR exhibits a weaker inversion tendency.

For a representative trap-evolution setting with $\eta = 0.15 \text{ ms}/^\circ\text{C}$, Fig. 11 anchors the turning temperature with consistent internal signatures. Both E_x and E_c exhibit significantly larger variations from -55°C to 30°C than from 30°C to 125°C , indicating that the primary barrier and field reconstruction occurs near the cold end and that further heating provides diminishing incremental driving force. Mobility degradation strengthens monotonically with temperature and contributes an additional read-domain shift under a constant-current criterion, especially in high states. NetTrappedCharge evolution further shows that the low-density, weakly bound component is preferentially depleted, leaving a progressively deeper-trap-dominated configuration.

Overall, TCAD reproduces the regime transition, its η -dependent trajectory sensitivity, and the deep/shallow occupancy contrast between HPLR and LPHR, providing a consistent physical basis for UCL and high-state OCH under a unified-slope linear TC scheme.

Building on the extracted V_{TH} distributions and the TCAD-supported mechanism insights, we summarize three actionable guidelines for improving wide-temperature read robustness under on-chip TC in charge-trapping 3-D NAND.

- 1) *TC Calibration Focus and Programming Temperature:* The implemented linear on-chip TC is primarily tuned for the nominal operating window, which makes end-point mismatch more likely at extreme temperatures. For the tested product, low-temperature programming is preferred when feasible, while high-temperature programming increases the risk of read-window loss and distribution compression in high program states under TC.
- 2) *Physical Origin of Endpoint Mismatch:* At the low temperature, trap-level occupancy and related energy terms vary rapidly with temperature, so a single global slope cannot fully cancel the steep response, leading to UCL. At high temperature, PF emission together with stronger mobility degradation in high states amplifies the intrinsic state dependence of the mean response, making a unified-slope correction prone to OCH.
- 3) *Trajectory and Trap-Evolution Effects:* Cross-temperature trajectories reshape the deep/shallow trap occupancy and introduce a trajectory-dependent sensitivity of the turning behavior to the trap-evolution setting controlled by η . This explains why HPLR is more susceptible to evident high-state OCH and inversion, whereas LPHR tends to show a weaker anomaly.

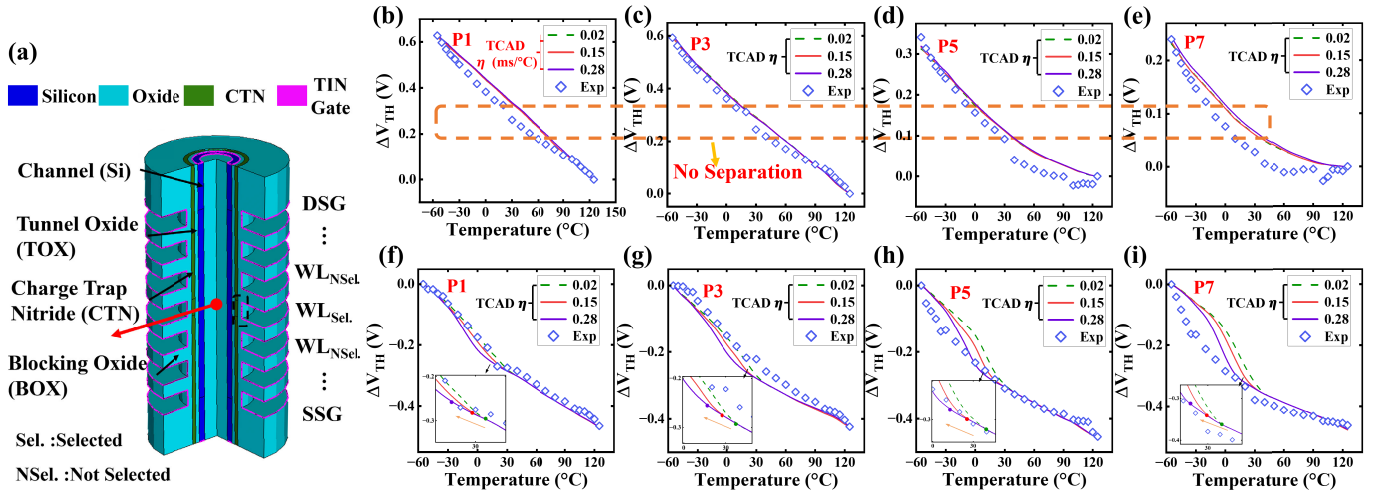


Fig. 10. TCAD structure and simulated mean $\Delta V_{TH}(T)$ for representative TLC states under the without TC condition. (a) Schematic of the simulated vertical-channel charge-trapping 3-D NAND structure. (b)–(e) P1, P3, P5, and P7 under HPLR, respectively. (f)–(i) P1, P3, P5, and P7 under LPHR, respectively. Multiple η settings are used to probe the trajectory-dependent sensitivity of the turning behavior to the effective trap-evolution time scale, rather than to reproduce a package-level thermal ramp.

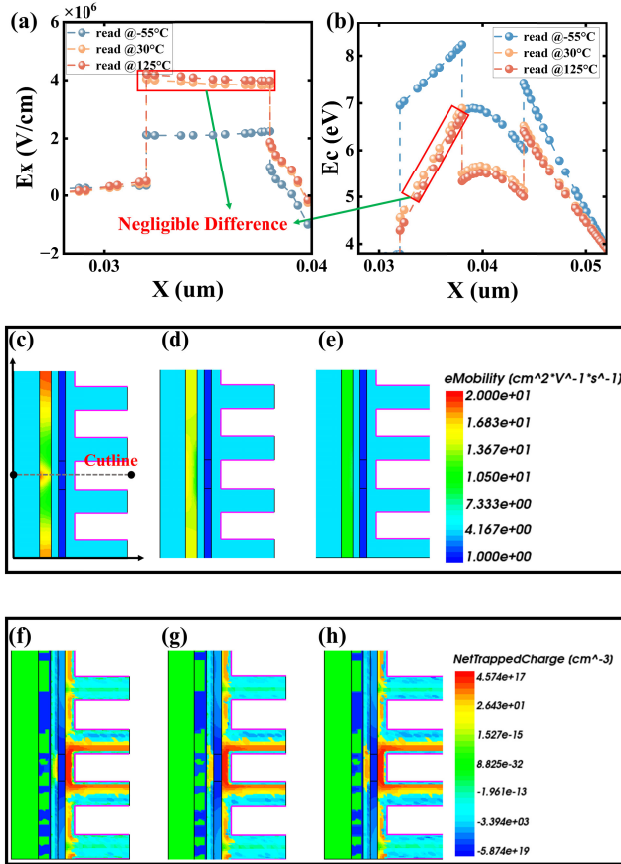


Fig. 11. Internal TCAD signatures at $\eta = 0.15$ ms/°C for interpreting the turning behavior. (a) Vertical electric field E_x at -55 °C, 30 °C, and 125 °C. (b) Conduction-band profile E_c at -55 °C, 30 °C, and 125 °C. (c)–(e) Channel electron mobility distributions at -55 °C, 30 °C, and 125 °C. (f)–(h) NetTrappedCharge distributions at -55 °C, 30 °C, and 125 °C.

IV. CONCLUSION

This work presents a distribution-level assessment of charge-trapping 3-D NAND from -55 °C to 125 °C and

identifies a systematic two-end mismatch of conventional unified-slope on-chip linear TC, manifested as UCL and OCH. The impact concentrates in high program states, where OCH compresses the effective read window and can induce a mean-response inversion under the HPLR trajectory. Sentaurus TCAD reproduces the regime transition and its trajectory dependence, providing physical support for the observed endpoint mismatch and the stronger high-state vulnerability. These findings indicate that state-insensitive linear TC is insufficient for robust readability at temperature extremes and motivate state-aware calibration for future controllers. In conclusion, the unraveled state-dependent high- and low-temperature discrepancies highlight the inadequacy of conventional unified-slope TC. These insights provide critical guidelines for developing advanced, state-aware TC algorithms in future 3-D NAND controllers intended for harsh-environment applications.

REFERENCES

- [1] N. Righetti and G. Puzzilli, "2D vs 3D NAND technology: Reliability benchmark," in *Proc. IEEE Int. Integr. Rel. Workshop (IIRW)*, Oct. 2017, pp. 1–6, doi: [10.1109/IIRW.2017.8361235](https://doi.org/10.1109/IIRW.2017.8361235).
- [2] D. Resnati et al., "Characterization and modeling of temperature effects in 3-D NAND flash arrays—Part I: Polysilicon-induced variability," *IEEE Trans. Electron Devices*, vol. 65, no. 8, pp. 3199–3206, Aug. 2018, doi: [10.1109/TED.2018.2838524](https://doi.org/10.1109/TED.2018.2838524).
- [3] G. Nicosia et al., "Characterization and modeling of temperature effects in 3-D NAND flash arrays—Part II: Random telegraph noise," *IEEE Trans. Electron Devices*, vol. 65, no. 8, pp. 3207–3213, Aug. 2018, doi: [10.1109/TED.2018.2839904](https://doi.org/10.1109/TED.2018.2839904).
- [4] C. Zhao et al., "Investigation of threshold voltage distribution temperature dependence in 3D NAND flash," *IEEE Electron Device Lett.*, vol. 40, no. 2, pp. 204–207, Feb. 2019, doi: [10.1109/LED.2018.2886345](https://doi.org/10.1109/LED.2018.2886345).
- [5] D. Wu, H. You, X. Wang, S. Zhong, and Q. Sun, "Experimental investigation of threshold voltage temperature effect during cross-temperature write-read operations in 3-D NAND flash," *IEEE J. Electron Devices Soc.*, vol. 9, pp. 22–26, 2021, doi: [10.1109/JEDS.2020.3035648](https://doi.org/10.1109/JEDS.2020.3035648).
- [6] M. Pesic et al., "Insights into device and material origins and physical mechanisms behind cross temperature in 3D NAND," in *Proc. IEEE Int. Rel. Phys. Symp. (IRPS)*, Mar. 2023, pp. 1–8, doi: [10.1109/IRPS48203.2023.10117898](https://doi.org/10.1109/IRPS48203.2023.10117898).

- [7] S. Choi et al., "A cell current compensation scheme for 3D NAND FLASH memory," in *Proc. IEEE Asian Solid-State Circuits Conf. (A-SSCC)*, Nov. 2015, pp. 1–4, doi: [10.1109/ASSCC.2015.7387432](https://doi.org/10.1109/ASSCC.2015.7387432).
- [8] C. Zambelli, E. Ferro, L. Crippa, R. Micheloni, and P. Olivo, "Dynamic VTH tracking for cross-temperature suppression in 3D-TLC NAND flash," in *Proc. IEEE Int. Integr. Rel. Workshop (IIRW)*, Oct. 2019, pp. 1–4, doi: [10.1109/IIRW47491.2019.8989886](https://doi.org/10.1109/IIRW47491.2019.8989886).
- [9] Y. Wang, J. Huang, J. Yang, and T. Li, "A temperature-aware reliability enhancement strategy for 3-D charge-trap flash memory," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 38, no. 2, pp. 234–244, Feb. 2019, doi: [10.1109/TCAD.2018.2808227](https://doi.org/10.1109/TCAD.2018.2808227).
- [10] C. Zambelli, L. Crippa, R. Micheloni, and P. Olivo, "Cross-temperature effects of program and read operations in 2D and 3D NAND flash memories," in *Proc. Int. Integr. Rel. Workshop (IIRW)*, Oct. 2018, pp. 1–4, doi: [10.1109/IIRW.2018.8727102](https://doi.org/10.1109/IIRW.2018.8727102).
- [11] Y. Guo, K. Xie, X. Fang, X. Zhan, J. Wu, and J. Chen, "Cross-temperature reliabilities in TLC 3D NAND flash memory: Characterization and solution," in *Proc. 7th IEEE Electron Devices Technol. Manuf. Conf. (EDTM)*, Mar. 2023, pp. 1–3, doi: [10.1109/EDTM55494.2023.10103107](https://doi.org/10.1109/EDTM55494.2023.10103107).
- [12] M. A. Kumar and B. Ray, "Cross-temperature reliability of 3D NAND: Cell-to-cell variability analysis and countermeasure," in *Proc. IEEE Int. Rel. Phys. Symp. (IRPS)*, Apr. 2024, pp. P25.MR-1–P25.MR-7, doi: [10.1109/irps48228.2024.10529488](https://doi.org/10.1109/irps48228.2024.10529488).
- [13] B. Choi et al., "Comprehensive evaluation of early retention (fast charge loss within a few seconds) characteristics in tube-type 3-D NAND flash memory," in *Proc. IEEE Symp. VLSI Technol.*, Jun. 2016, pp. 1–2, doi: [10.1109/VLSIT.2016.7573385](https://doi.org/10.1109/VLSIT.2016.7573385).
- [14] C. Woo et al., "Modeling of charge loss mechanisms during the short term retention operation in 3-D NAND flash memories," in *Proc. Symp. VLSI Technol.*, Jun. 2019, pp. T214–T215, doi: [10.23919/VLSIT.2019.8776579](https://doi.org/10.23919/VLSIT.2019.8776579).
- [15] S. Xia et al., "Analysis and optimization of temporary read errors in 3D NAND flash memories," *IEEE Electron Device Lett.*, vol. 42, no. 6, pp. 820–823, Jun. 2021, doi: [10.1109/LED.2021.3073604](https://doi.org/10.1109/LED.2021.3073604).
- [16] C. Park, J. Kim, K. Nam, D. Kim, H. You, and R.-H. Baek, "A physical retention model valid for retarded cross-temperature instability analysis of 3-D NAND flash memory," *IEEE Trans. Electron Devices*, vol. 72, no. 12, pp. 6687–6692, Dec. 2025, doi: [10.1109/TED.2025.3621170](https://doi.org/10.1109/TED.2025.3621170).
- [17] K. Lee and H. Shin, "Investigation of retention characteristics for trap-assisted tunneling mechanism in sub 20-nm NAND flash memory," *IEEE Trans. Device Mater. Rel.*, vol. 17, no. 4, pp. 758–762, Dec. 2017, doi: [10.1109/TDMR.2017.2772046](https://doi.org/10.1109/TDMR.2017.2772046).
- [18] H. Jo and H. Shin, "Compact modeling of trap-assisted tunneling current in 3-D NAND flash memory," *IEEE Trans. Electron Devices*, vol. 72, no. 4, pp. 1745–1749, Apr. 2025, doi: [10.1109/TED.2025.3541603](https://doi.org/10.1109/TED.2025.3541603).
- [19] J. Park et al., "Decomposition of charge loss mechanisms in 3-D NAND flash memory: Impact of cell dimension via high-temperature long-term retention characteristics," *IEEE Trans. Electron Devices*, vol. 71, no. 10, pp. 6040–6048, Oct. 2024, doi: [10.1109/TED.2024.3449251](https://doi.org/10.1109/TED.2024.3449251).
- [20] G. Malavena, M. Giulianini, L. Chiavarone, A. S. Spinelli, and C. Monzio Compagnoni, "Analysis of high-temperature data retention in 3D floating-gate NAND flash memory arrays," *IEEE J. Electron Devices Soc.*, vol. 11, pp. 524–530, 2023, doi: [10.1109/JEDS.2023.3320722](https://doi.org/10.1109/JEDS.2023.3320722).